Faculty	**FET**		
Assessment Name	**END**	**Paper Code**	**CPN1/G25**
Module Name	**Programing using .Net**	**Module Code**	**C7-PN1-11**
Semester (E.g Jul-Dec-2021)	**Jul - Dec**	**Submission Mode** (E.g Turnitin, blackboard, hardcopy)	**Turnitin & Blackboard**
Total Marks	**100**	**Duration** **(E.g., mins/hours/days/weeks**	
Assessment Type (Eg: Written/Practical/Submission)	**Submission**	**Submission Date** (Applicable for submission-based exams only)	**13th Nov 2025**

Instructions

1. This project must be done by each student individually.
2. The use of AI tools in this assessment is **NOT** allowed.
3. Read and understand the project requirements and address them using C# Windows Forms concepts.
4. Use **Visual Studio** and the **C# language** for coding the project.
5. You must strictly adhere to the submission deadlines. The due date is **13th November 2025**. No late submissions will be accepted.
6. You are required to submit a Project Document that includes the Analysis, Design, and Implementation sections.
7. Plagiarism is a serious academic offense. Submitting plagiarized work can result in your marks being nullified. You must submit your report to Turnitin.com to check for plagiarism.
8. Submit your zipped source code to the submission link on Blackboard.

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1. Project Scenario

The new Botho University campus in Maseru requires a modern digital solution to manage its student clinic. To replace the current inefficient paper-based system, you are tasked with developing the **"Botho University Campus Clinic Management System."** This robust **C# Windows Forms application** will serve as the clinic's Electronic Health Record (EHR) and appointment scheduling backbone, aiming to improve patient care, operational efficiency, and data-driven health monitoring on campus.

2. System Modules, Functionalities, and Forms

The application must be a role-based system. Access to functionalities and data is strictly determined by the user's role after a secure login.

2.1 User Registration and Login

User registration is a controlled process managed exclusively by a Clinic Administrator to ensure system security and data integrity.

- **Expected Form: frmLogin**
  - **Purpose:** To securely authenticate all users.
  - **Controls:** TextBoxes for Username and Password (masked), a Login Button, and a Label for error messages.
  - **Logic:** Upon successful authentication, the system identifies the user's role and loads the appropriate dashboard.
- **Expected Form: frmUserRegistration (Admin Access Only)**
  - **Purpose:** To create new user accounts for students and clinic staff.
  - **Controls:** TextBoxes for Student/Staff ID, Full Name, Contact Details, and a ComboBox to assign a Role (Student, Healthcare Provider, Administrator).
  - **Logic:** The administrator creates the account with a default password, which the new user must change upon their first login.

2.2 Student (Patient) Module

This module provides students with simple, secure access to their personal health services.

- **Core Functionalities:**
  - Book and manage clinic appointments.
  - View a history of past consultations.
  - Receive health reminders from the clinic.
  - Manage their personal profile and password.
- **Expected Forms:**
  - **frmStudentDashboard:** The student's main screen, displaying their next **upcoming appointment** and a list of health reminders. Should contain navigation buttons to other student forms.
  - **frmBookAppointment:** An interactive form with a MonthCalendar to select a date, a ListBox

or similar control to show **available time slots**, and a TextBox for the reason for the visit.
    - **frmAppointmentHistory:** A read-only form with a DataGridView listing all past appointments, showing the date, provider, and reason/diagnosis.

2.3 Clinic Nurse/Doctor (Healthcare Provider) Module

This is the primary clinical interface for managing patient care and medical records.

- **Core Functionalities:**
    - View a daily schedule of appointments.
    - Securely access and update patient electronic health records.
    - Document consultation details, including vitals, notes, and diagnoses.
    - Issue and record prescriptions.
- **Expected Forms:**
    - **frmProviderDashboard:** The provider's workspace, featuring a DataGridView of **today's appointments**. Double-clicking a patient's name should open their consultation form.
    - **frmConsultation:** A comprehensive, tabbed form for managing a patient visit.
        - **Vitals & Notes Tab:** Fields for recording temperature, blood pressure, and detailed consultation notes/diagnoses.
        - **Medical History Tab:** A read-only DataGridView showing the patient's past visit summaries.
        - **Prescriptions Tab:** Controls to select medications, specify dosages, and add them to a prescription list for the current visit.
        - A master **Save Consultation** button is required.

2.4 Clinic Administrator Module

This is the high-level module for clinic operations, user management, and data oversight.

- **Core Functionalities:**
    - Monitor real-time clinic activity and campus health trends.
    - Manage user accounts for both students and staff.
    - Generate and export administrative reports.
- **Expected Forms:**
    - **frmAdminDashboard:** An overview screen with **KPI cards** ("Patients Seen Today," etc.) and a Chart control visualizing a key metric, such as "Common Ailments This Month."
    - **frmStaffManagement:** A form to view all staff accounts in a DataGridView, with options to **deactivate accounts** or **reset passwords**. This form should also provide access to the frmUserRegistration form.
    - **frmReporting:** A simple form with two DateTimePicker controls to select a date range and a button to **"Export to CSV,"** which generates a monthly clinic activity report.

3. Technical & Non-Functional Requirements

- **Technology Stack:** C# Windows Forms, ADO.NET, and SQL Server or MySQL.
- **Database Design:** The schema must be in at least **Third Normal Form (3NF)**. An Entity-Relationship Diagram (ERD) and 3NF justification are required in your report.
- **Exception Handling:** The application must use try-catch blocks to gracefully handle database connection issues, file I/O errors, and invalid user input without crashing.
- **Data Structures & Collections:** Demonstrate the effective use of C# collections (e.g., List<T>, Dictionary<T,V>) for managing data in memory.
- **Security & Privacy:** This is a critical requirement. Access must be strictly role-based. Passwords must be **hashed**, and all SQL queries must be **parameterized** to prevent SQL injection.

4. Database Schema Guide

Your database design should logically separate data into tables such as Users, Roles, Patients, Appointments, Consultations, and Prescriptions, using primary and foreign keys to establish relationships.

5. Submission Deliverables

Submit a single .zip file containing:

1. **Project Source Code:** The complete, compilable Visual Studio project.
2. **Database Script:** A .sql file to create the schema and insert sample data.
3. **Project Report:** A comprehensive .pdf report detailing the project's design, implementation, and evaluation.

6. Assessment & Detailed Marking Rubric (Total: 100 Marks)

Category	Criteria	Marks
Database Design	• ERD is clear and logical. • Schema is correctly normalized to **3NF** with clear justification in the report. • SQL script runs without errors.	**15**
Core Functionality	• All requirements for all three user roles are implemented correctly. • Role-based login and permissions are strictly enforced. • CRUD operations on appointments and records are fully functional.	**20**
Advanced Technical Implementation	• **Exception Handling:** Robust try-catch blocks are used for database, file I/O, and input validation. • **Collections:** Effective use of collections is demonstrated. • **File System:** "Export to CSV" feature works correctly. • **Security:** Passwords are hashed and all SQL	**25**

	queries are parameterized.	
Code Quality & GUI	• Code is well-structured, commented, and follows best practices. • The Windows Forms GUI is professional, intuitive, and user-friendly.	**10**
Project Report	• Detailed discussion of requirements, design, and implementation. • Includes clear diagrams and well-commented code snippets. • Professional formatting and grammar.	**15**
Final Demonstration & Q&A	• Clear and logical presentation of all project features. • Ability to confidently answer questions about the code, design choices, and security considerations.	**15**
TOTAL		**100**